

BaseRate

Screening statistics, explained through people you can count.

THE PROBLEM AND AUDIENCE

Students and health-literacy learners can confuse detection among cases with the chance of a condition among positive results. BaseRate makes the denominator visible through fictional populations, counterfactual experiments and teach-back feedback.

THE WORKING EXPERIENCE

Edit prevalence, sensitivity and specificity. Switch between 1,000 population squares and a frequency tree. Pin experiment A, vary B and explore the prevalence curve. Inspect conditional repeat-test assumptions. Complete three predict/reveal/denominator cases and export a numeric notebook. Explain the result to an on-device AI coach with inspectable, source-linked feedback.

SAME TEST, DIFFERENT STARTING POPULATION

90% sensitivity + 95% specificity, in 1,000 fictional people

1% prevalence: $9 / (9 + 49.5) = 15.4\%$ PPV. At 20%: **81.8%**.

AI AND PRIVACY

A Multinomial Naive Bayes classifier learns word and adjacent-word features from 100 synthetic English explanations. Five reasoning labels select fixed educational responses. Coverage and uncalibrated vote separation can trigger uncertainty. Arithmetic runs separately. No patient data, analytics, remote inference or stored learner text; exports exclude explanations.

WHAT WAS VERIFIED

Four reproducible suites pass: 336 base arithmetic/rounding combinations, 2,525 curve points, 405 conditional repeat-test combinations, lesson equations, invalid inputs, immutable snapshots, tiny-count display and text-free exports. Browser checks cover connected flows and a 390-pixel layout. Independent code-review findings were corrected.

HONEST LIMITS AND NEXT VALIDATION

A separate 30-example synthetic holdout gives 29 correct raw classifications; uncertainty checks accept 28 labels, 27 correct. One correct sensitivity distinction is misread. These are development checks, not evidence of learning benefit or clinical accuracy. All rates are fictional; English only; no personal diagnosis or test recommendation. Next: educator review, independent learner data and a comprehension study.

PUBLIC PROJECT AND AUTHORSHIP

nicholas-afk.github.io/baserate | GitHub source, data and tests. Static HTML/CSS/JavaScript, standard-library Python trainer. Code, synthetic data and submission materials use extensive AI assistance; demo narration is the entrant's recording. The entrant must understand and explain the code.

Sources: Hoffrage et al., 2015, original natural-frequency study; NCI, Cancer Screening Overview. Sources inform content, not this prototype's effectiveness.